

Disease models

Sepsis

8 state variables
ODE: pathogen, immune,
organ damage

ARDS

6 state variables
SDE: alveolar flooding
and compliance

ACS

8 state variables
ODE: troponin, two
compartments

parameters θ and initial conditions

Patient digital twin core

State vector $x(t)$, n dimensions · horizon 24 h · $\Delta t = 1$ h, 25 observations per twin
Integration: explicit forward Euler for all three disease models
Reproducibility: deterministic seeding (seed 42) across $N = 1,000$ twins
Intervention response: a physiologically grounded Δx applied after integration

true state $x(t)$

Measurement process simulator

Observation model: $o_i(t) = s_i(t) + \varepsilon_i(t)$ when measured, otherwise missing
Noise: $\varepsilon_i \sim N(0, \sigma_i^2)$, drawn independently at each observation time
Observation times: every integration step, hourly across the 24 h horizon

observations $o(t)$

Missing data

MCAR, 20%,
used in results;
MAR/MNAR ship only

Noise

Gaussian, $\sigma \times 2$,
used in results;
Laplacian, $t(3)$ ship

Temporal mask

Gaps of 1–4 h;
ships but is not
used in any result

Conflict

Discordant biomarker
trends; ships but is
not used in any result

perturbed observations

Logistic
regression

Random
forest

Gradient
boosting

XGBoost

LSTM

TCN

Metrics: AUROC · accuracy · TCS · TCE · DIR · counterfactual regret · DBRS · BEWS

Twin core

Measurement

Perturbation

Evaluation